

Findings of Record — Full-Sample Two-Grid Analysis Run of August 17, 2026

Run: replication_20260817_015626 · **Stack:** v0.9.86 · **Sample:** 66 sessions (37 volatile, 25 benchmark, 4 MWCB), 2017-06-26 – 2026-06-05 · **Grids:** 1 s and 10 ms (books pulled once at 10 ms; 1 s frames derived) · **Prepared:** August 18, 2026, from the shipped analysis bundle (132 files). Every headline and data-quality figure cited below was independently re-read against its source table before inclusion; one figure in the source memoranda was corrected in the process (§ I.2).

I. Provenance and Completeness

I.1 — Stage record. All driver stages completed except one. Extraction: 66 of 66 sessions usable (28,409.8 s, 4 workers). STAGE 1: all correctness gates passed. The 1 s analysis ran 13 of 13 stages ok (3,296.9 s); the 10 ms analysis ran its designed 7-stage subset ok (18,678.2 s); flow tables shipped at both grids; copula, geometry, DCC, IRF/FEVD, panel, robustness, and legacy exhibits shipped at 1 s by design; the horizon ladder shipped from the 10 ms frames. The sole failure: **the 10 ms Table 9**, killed during the Pearson-column rolling estimation (peak RSS $\approx$ 292.6 GiB); no 10ms/table9 output exists. The Markov-switching regime analysis, pending in the August 17 memorandum, completed and is on the record (§ IV.11). (*replication.log*; *MANIFEST.md*)

I.2 — Correction to the August 17 memorandum. The memorandum's stale-session regime tally ("~6 benchmark, 19 volatile" in § II.A; "11 benchmark, 14 volatile" in § V) is wrong in both places. The 25 sessions failing the ES ladder invariant at 10 ms comprise **16 benchmark and 9 volatile days** (0 MWCB). As proportions of regime: **64 % of benchmark sessions are stale versus 24 % of volatile sessions**. Every directional argument built on the memorandum's tally inverts accordingly — see § II.1. (*qc/qc_frames_10ms.txt*)

I.3 — What did not run. No table9_lag_band_* file exists at either grid: the lag-band robustness harness was not invoked, so under the bundle's own convention no Table 9 cell is certified lag-robust from this run. The 10 ms leg of the Epps pair (Table 9) is absent per I.1.

II. Data-Quality Findings

D-1 — ES ladder staleness is severe, early-era, and concentrated in the benchmark regime. At 10 ms, 25 of 66 sessions have the ES top of book unchanged on 99.0–99.9 % of snapshots (range 0.9904–0.9994); a further 38 sessions sit between 0.90 and 0.99; the sample minimum is 0.885 (2020-03-18). No session in 2022, 2025, or 2026 exceeds 0.99. At 1 s the worst four sessions (0.92–0.97) are all 2017–2018 benchmark days, including 2017-11-28, the replacement control. The regime asymmetry (64 % of benchmark vs 24 % of volatile sessions stale at the 0.99 threshold) means any staleness-induced bias loads on the **benchmark** side of every regime contrast. The v0.9.87 measured bias curve (planted truth through the real Tier-1 pipeline) finds **attenuation** — bias -0.90 z at 85 % staleness, monotone — under the carry-forward mechanism; under that mechanism the benchmark flow correlation is understated and the volatile-minus-benchmark gap is conservative. The refresh-clustering channel (which would push the other way) is not in the simulated curve and remains open. (*qc/qc_frames_10ms.txt*; *qc/qc_frames.txt*)

D-2 — The run estimated at 10 ms on sessions its own QC forbids. *qc_frames_10ms.txt* marks the 25 stale sessions `ok=False` with the standing instruction "Do NOT estimate on a session that violates its own invariant"; extraction recorded them `DEGRADED` under `qc_action=warn`; STAGE 6b, the 10 ms flow stage, and the horizon ladder estimated on all 66. Every 10 ms headline therefore includes the stale sessions. Disposition options: the P0 clean-subsample rerun, the `es_stale_frac` dose-response covariate (shipped in v0.9.87), or per-exhibit exclusion — but the current record should be read with D-1 in hand. (*qc/qc_frames_10ms.txt*; *frames/extract_report.txt*; *replication.log*)

D-3 — One ladder-integrity defect. 2017-06-26 violates ES ladder monotonicity on 8 of 2,340,001 snapshots (ES_ladder_mono = 0.999997); the QC note suspects a level-column mapping or partially applied price scale. All other sessions are exactly 1.0. The same session is the sample's session0 illustration in several single-session exhibits (§ V, E-12). (*qc/qc_frames_10ms.txt*)

D-4 — Masking is verified in the estimates. MWCB halt rows (900–901 SPY / 819–847 ES at 1 s per day; ×100 at 10 ms) and the 2021-11-26 post-13:00 segment (10,801 rows at 1 s; 1,080,001 at 10 ms) are QC-excluded and estimator-masked; the mediation panel's 24,397 rows equal the nominal bar count minus masked bars; SPY crossed_open never exceeds 0.17 %. ES books never cross at either grid. (*qc/qc_frames.txt; replication.log; flow tables*)

D-5 — 2020-12-11 contract identity, resolved. The extraction log shows the activity rule extracted ESZ0 (75.8 % of two-contract volume) over calendar front ESH1; the QC table's roll_rule_ok=N0 and ES_front_vol=24.2 % score the **calendar front**, not the extracted leg, and the QC caption ("calendar rule picked the minority contract") is written for the un-overridden case. The frame's median ES bid (3651.00 against SPY 365.15) is consistent with ESZ0. The v0.9.87 QC table adds an es_contract column so the ambiguity cannot recur. The remaining roll facts stand: measured splits 2020-03-18 (69.5 %) and 2024-12-18 (64.2 %); cleared-volume-table fills for the two unmeasured roll-window sessions (2020-03-16: calendar front 24.7 % of prior-session cleared volume; 2022-06-13: 46.6 %), reported and not acted on. (*replication.log; qc/qc_frames.txt*)

D-6 — SSR. Rule 201 was in force for 100 % of 2020-03-16, 2021-11-26, and 2022-02-10; the SSR state of the six 2017 sessions is unknown (source silent — absence is not "unrestricted"). Table 5's MWCB panel is reported with and without its SSR member. (*qc files; replication.log STAGE 4*)

D-7 — Unequal bootstrap resolution across grids. The 1 s mediation CIs rest on 499 cluster-bootstrap draws; the 10 ms mediation CIs on 199, against a configured n_boot = 499. The MS-regime bootstrap runs 99 draws per stage everywhere, so its "median p = 0.01" cells are a resolution floor, not an estimate. (*flow_corr_mediation_.md; flow_corr_ms_regimes_.csv*)

D-8 — Dead columns. Equity LULD columns are all-NaN on all 66 sessions; ES_luld_known is 0 % on 65 of 66 (4 % on 2021-11-26) despite the QC narrative asserting futures limits "ARE reported." Neither is usable as a control on this record. (*qc files*)

III. Method Findings

M-1 — Clustering is worth a factor of seventy. SPY_ret ~ ES_OFI over 1,529,994 observations: pooled-iid t = 386.87; day-clustered t = 5.56 (wild-bootstrap p = 0.001). The 69.5× deflation is the single most consequential inference fact in the run and governs the reading of every pooled statistic below. (*legacy__inference_iid_vs_clustered.csv*)

M-2 — The Table 5 z-statistics are bar-count artifacts. log_OR_z of 159.8 / 214.3 / 77.7 / 64.2 across the four panels are Woolf statistics on pooled bars; the z ordering tracks panel size, not effect size (the largest log OR, MWCB 1.470, has near the smallest z). The v0.9.87 day-clustered inference (per-day count matrices; day bootstrap; within-pair contrast; MWCB jackknife) replaces them; on a planted serially-dependent DGP the Woolf z overstates the day-clustered z by ~21×. (*nulls/table5_corrected_null.csv*)

M-3 — The lag-order record is untidy but internally explicable. STAGE 4c resolved p = 10 (BIC, RealBar frame, 1 s); the Table 9 frames' own BIC selected p = 5; the per-day modal selection is p = 1 on 92–95 % of days; AIC never attains an interior minimum (still falling at the pmax = 30 bound — the footnote-17 artifact seen from below); the VECM stages use per-grid heuristics (5 at 1 s, 60 at 10 ms). Three different "BIC lags" appear in the record with no reconciling exhibit. The v0.9.87 local-projection Table 9 counterpart is lag-order-free and renders the dispute moot for the IRF claims it covers. (*replication.log; table9_lag_ic_informational_w100.csv*)

M-4 — Identification: the Rigobon column is unusable, by its own diagnostic. Cholesky (futures-first) gives contemporaneous $SPY \leftarrow ES = 0.892$; the heteroskedasticity-ID column reverses it ($-0.090 / +0.976$) but its var-ratio spread is 0.066 (~common-scale, unidentified) and the printed verdict is "NO — do not quote the het-ID column." The Cholesky ordering stands by construction, not by test. Note an annotation defect: the eigenvalue-gap row (0.438) clears its own stated >0.10 threshold while the verdict is NO; the verdict is keyed to the var-ratio spread. (*legacy__identification_Cholesky_vs_Rigobon.csv*)

M-5 — FEVD vs GFEVD. With measured OFI-innovation correlation 0.705 (benchmark) / 0.788 (volatile), the diagonal FEVD is invalid; the GFEVD attributes 45.7 % of SPY-return variance to ES flow (benchmark) and 50.0 % (volatile) versus 27.9 % / 50.5 % under the diagonal. The shipped FEVD/GFEVD CSVs are the **benchmark-regime** tables; the volatile-regime decompositions exist only inline in report.md (§ V, E-9). (*irf__fevd.csv*; *irf__gfevd.csv*; *irf__ofi_innovation_corr.csv*; 1s report.md)

M-6 — DCC is a boundary estimate. $a + b = 0.99990$; mean conditional ρ 0.811 vs realized correlation 0.923. Per convention the realized correlation is the quotable level; DCC's near-zero IRF responses in Table 9 (§ IV.3) should be read against this. (*dcc.csv*)

M-7 — CS is the fragile statistic, quantified. Mean CS_ES drifts $+0.071$ (22 %) across VECM lags $3 \rightarrow 20$ while IS moves -0.014 ; single-day CS swings up to 0.78 between fixed and estimated cointegrating beta (2021-11-26: $0.024 \rightarrow 0.801$) while the largest IS swing is 0.127; nine (1 s) / ten (10 ms) low- κ days carry the extreme CS values; excluding them nearly eliminates the 1 s CS regime gap. IS is the primary metric throughout. (*robustness__lag_sensitivity.csv*; *robustness__beta_fixed_vs_estimated.csv*; *information_shares__per_day.csv*)

M-8 — Two regime taxonomies coexist. Copula and Table 5 exhibits break MWCB out (volatile = 37); the per-day analysis tables, cross-impact, horizon, geometry, and state-conditioning tables fold MWCB into volatile (= 41). Documented in RUN_NOTES as intended, but every cross-family "volatile" comparison mixes day sets, and the MWCB days are extreme on most measures. (*RUN_NOTES.md*; *per-day tables*)

M-9 — Era drift is real and untested. IS_mid_ES rises monotonically across every market-era flag at both grids (e.g., pre/post T+1: $0.447 \rightarrow 0.492$ at 1 s; $0.437 \rightarrow 0.565$ at 10 ms), but era cells are regime-imbalanced (post-tick-2025 contains zero benchmark days) and κ falls simultaneously; no era test ships. Book-geometry levels drift similarly (2017 $SPY \tau_{bid} \approx 0.058$ vs 2024–26 ≈ 0.13 –0.21). Cross-era level comparisons are confounded without the within-pair design. (*information_shares__per_day.csv*; *book_geometry_per_day_1s_L10.csv*)

IV. Findings on the Substantive Questions

F-1 — Tandem flow coupling orders monotonically with stress, on both grids. Tier-1 OFI-innovation correlation: benchmark $0.729 < \text{volatile } 0.818 < \text{MWCB } 0.870$ at 1 s (volatile–benchmark $dz +0.226$, permutation $p = 0.0008$); benchmark $0.266 < \text{volatile } 0.292 < \text{MWCB } 0.355$ at 10 ms ($dz +0.028$, $p = 0.0327$). This is the only flow exhibit that agrees across grids in direction and significance, and it is the run's most robust substantive result. The staleness bias curve (D-1) says the measured mechanism biases it conservatively. (*flow_corr_tier1_.csv*)*

F-2 — The asymmetry reversal is a 1 s result; it does not replicate at 10 ms. At 1 s: benchmark sell-side tighter ($dz +0.047$, sign-flip $p = 0.0045$); volatile buy-side tighter ($dz -0.090$, $p < 5 \times 10^{-5}$); MWCB symmetric ($p = 0.62$). At 10 ms every group's gap is an order of magnitude smaller and non-significant (volatile $dz -0.0023$, $p = 0.47$). The gap-of-gaps (-0.137 at 1 s) is itself untested in the shipped tables. The reversal remains the most novel candidate finding, but the record supports it only at the 1 s aggregation, and the SSR/direction-conditioning checks (memorandum P1; §VI) remain outstanding. (*flow_corr_asymmetry_.csv*)*

F-3 — Mediation: the mediator link is rock-solid; the decomposition is grid-dependent. The within-bar mediator coefficient ($\Delta z_{flow} \rightarrow \Delta z_{ret}$) is 67.0 ($t \approx 17.8$) at 1 s and 75.3 at 10 ms — the strongest single coefficient in the run.

The suppression structure (direct +5.71 vs indirect -6.84, total ≈ 0) holds at 1 s, where the indirect **level** CI [-41.9, -3.14] excludes zero; the share statistic (605 %) is a ratio to a near-zero denominator and should not be quoted. At 10 ms the total is significant (-1.076***), the direct keeps its sign, and the indirect CI includes zero. The two grids agree that RV_ES suppresses flow innovations (Eq A negative both grids) and disagree on the rest. RV_SPY's indirect channel is never reported despite significant Eq-A coefficients on both grids. (*flow_corr_mediation_.csv/.md*)*

F-4 — Lead-lag: ES leads by $\approx$ one 20 ms step — at the grid that can see it, and net of staleness. At 10 ms, raw HRY $\theta = +0.020$ s on 62 of 66 days (64 ES-leading; sign-flip $p = 0.0005$) and the co-jump lead test gives ES-first on 65 of 66 days (mean lead share 0.358, $p = 5 \times 10^{-5}$; pooled 308,921 ES-first vs 135,777 SPY-first). At 1 s neither test resolves anything ($p = 0.52 / 0.96$) — a resolution artifact, not a contradiction. Two integrity notes: (i) the co-jump ES-lead share correlates **positively** with ES update counts (Spearman +0.57), i.e. the result is weakest on the stale sessions and is not a staleness artifact; the two extreme raw- θ days (2017-06-26, 2017-10-05) are the two stalest sessions and should be excluded from θ magnitudes. (ii) The pre-averaged θ column reverses sign on nearly every day (62/66 negative at 10 ms) and ships untested — the raw-vs-preavg contradiction is unadjudicated on this record (§ V, E-2). (*lead_lag, jumps__cojump tables*)

F-5 — Information shares: parity-to-SPY tilt on average; the regime shift is not era-robust. Mean IS_mid_ES ≈ 0.457 (1 s) / 0.466 (10 ms); CS_ES 0.323 / 0.381. The only sub-0.05 regime result is the 1 s free-permutation IS test (diff +0.032, $p = 0.032$); the era-robust within-pair test was run only on CS, where it reverses sign at 1 s (mean pair diff -0.002, $p = 0.976$) and is positive but insignificant at 10 ms (+0.038, $p = 0.132$). **No within-pair IS test exists in this run** — the single significant regime result has no era-robust counterpart. v0.9.87 adds the within-pair and ex-straddle tests on IS_mid_ES (regime_p_IS_within_pair), so the next run carries the era-robust verdict on the primary metric. The pooled panel-VECM interaction is significant on the ES equation at both grids ($p = 0.0099 / 0.0297$) but is pooled, not day-level, inference. (*information_shares__regime_test.csv; summary.json*)*

F-6 — Tables 5 and 7. Corner concentration: PCMOF observed/null ratio rises 1.243 $\rightarrow$ 1.280 $\rightarrow$ 1.372 (baseline $\rightarrow$ volatile $\rightarrow$ MWCB) and NCMOF falls 0.763 $\rightarrow$ 0.730 $\rightarrow$ 0.659; log OR 0.977 $\rightarrow$ 1.124 $\rightarrow$ 1.470 (1.413 ex-SSR); corner asymmetry ≈ -0.02 in every panel (no SSR fingerprint at the pooled level). Table 7: the NCMOF excess over the frequency-matched null collapses from 33.5 $\times$ (1 s bars) to 1.32 $\times$ (10 ms) to 0.97 (action time) — the published aggregation story reproduces on this sample. Inference for both awaits the v0.9.87 day-clustered machinery (M-2); Table 7 ships no uncertainty at all. (*nulls/table5_corrected_null.csv; nulls/table7_frequency_matched_null.csv*)

F-7 — Copulas: three exhibits, three stories, one significant contrast. Returns: tail-dependence **level** rises with stress (median λ 0.531 benchmark $\rightarrow$ 0.685 volatile) but no crash-vs-rally asymmetry anywhere (all sign-flip $p \geq 0.125$); benchmark days select the symmetric t on 19/25 — which forces $\lambda_L = \lambda_U$, so the parametric asymmetry test is constrained exactly where it matters (remedied by the v0.9.87 nonparametric columns). Flows: volatile days select Frank (zero tail dependence) on 32/37 — extreme flow surprises **decouple** under stress while returns tighten. Liquidity split: crash-tail dependence is higher in thin-book halves on volatile days, $\Delta\lambda_L = +0.076$, sign-flip $p = 5 \times 10^{-5}$ — the only significant copula contrast in the bundle, and a direct book-state $\rightarrow$ tail-risk link. The "median p(upper-tail LR) = 0.0" column is degenerate in all eight groups and contradicts the Frank selections; do not quote it (§ V, E-6). (*1s/copula*)*

F-8 — The horizon ladder localizes cross-impact transmission at 250 ms. SPY-from-ES cross-impact rises monotonically 0.038 (10 ms) $\rightarrow$ 0.370 (1 s); the per-day half-impact horizon lands at 250 ms on 39 of 66 days (median 250 ms in both regimes), but volatile days reach half-impact by 100 ms on 19/41 versus 3/25 benchmark — transmission is faster under stress. The reverse direction (ES-from-SPY) has no defined half-impact horizon (non-monotone, negative benchmark medians); the summary's mean half-life column is outlier-dominated (2024-12-18 contributes 1,000–3,400 s rows) — use the monotone per-day medians (1.7 $\rightarrow$ 17.4 s). (*horizon*)*

F-9 — Cross-impact is one-directional except during MWCB weeks. SPY-return-on-ES-flow is positive and significant on 66 of 66 days (benchmark mean 0.306, volatile 0.409); ES-return-on-SPY-flow is ≈ 0 in benchmark (median -0.009; significantly negative on 14/25 days) and small-positive volatile (0.050) — but switches on,

significantly, on all four MWCB days (0.073–0.178, t 3.1–6.7). The pooled "cross" row in the regime summary (0.156 / 0.229) masks this ~50-fold direction asymmetry. (*cross_impact_panel.csv*)

F-10 — Jumps: a larger share of QV on benchmark days. Common-factor jump fraction 0.350 benchmark vs 0.186 volatile at 1 s (truncation; same ordering per asset and under LM) — volatile days are volatile through the continuous component, not through a higher jump share. The ES share of the jump component is grid-stable (≈ 0.50) while its continuous-component share is grid-dependent (0.43 $\rightarrow$ 0.24). 10 ms jump fractions (0.78) are staleness-inflated and method-confounded; quote the co-jump lead ratios instead. (*jumps.csv*)*

F-11 — MS regimes (completed): multi-regime structure is near-universal; alignment with the a-priori labels is qualitative only. 64/66 days multi-regime at 1 s (58/66 at 10 ms); top-regime occupancy orders with the design: benchmark 0.231 < volatile 0.387 < MWCB 0.729. Benchmark low-coupling states are significantly more persistent at 1 s (median durations 156/36 bars, $p = 0.00075$) — not replicated at 10 ms ($p = 0.98$). Pooled entry-on-selling (entry return -2.76 bps, $p = 0.020$) is 1 s-only. No formal state-alignment statistic (confusion matrix, concordance) ships; every "median $p(1v2) = 0.01$ " cell is the 99-draw floor. (*flow_corr_ms_regimes_.csv*)*

F-12 — The flow-correlation $\rightarrow$ price-discovery link (pd_link) disagrees across grids in sign. All six 10 ms coefficients positive and significant (IS row +0.0151 to +0.0165**); all 1 s coefficients null with negative CS points. The exhibit conflates grid with specification (VECM lag 5 vs 60). On this record the "tandem flow moves price discovery toward ES" claim is 10 ms-only. (*flow_corr_pd_link_.csv*)*

F-13 — Book geometry: the redundancy verdicts hold on real data. τ is rank-equivalent to the Herfindahl on every one of 132 asset-days (median Spearman 0.972); incremental R^2 over the covering set: median 0.002 (τ), 0.001 (DWC). Both ship as diagnostics, not regressors, exactly as designed. (*book_geometry_per_day_1s_L10.csv*)

F-14 — ECM-SDE: state-dependence of adjustment speed is significant only at 10 ms, and the exhibit carries a unit defect. a_1 t -statistics: 0.30 (1 s mid) vs 3.68 (10 ms mid) / 7.26 (10 ms microprice); microprice raises the implied median ES share at both grids (0.176 $\rightarrow$ 0.308; 0.569 $\rightarrow$ 0.646). The *half_life_s* column is computed in **bar units** at both grids — at 10 ms the printed 137.5 "seconds" is 1.375 s of clock time. The cross-grid ECM-SDE leadership reversal (SPY-dominant at 1 s, ES-dominant at 10 ms) is unreconciled with the grid-stable Hasbrouck IS. (*ecm_sde.csv*)*

V. Internal Inconsistencies Requiring Disposition

E-1 · The 1 s co-jump lead test labels the leader "ES" while its own per-day table shows 15 of 30 informative days leaning SPY and pooled counts 380 SPY-first vs 180 ES-first (mean +0.006, $p = 0.96$); the label follows the sign of a statistically null mean. The summary.json *cojump_leader* field inherits it. E-2 · *theta_preavg_s* is opposite-signed to raw θ on essentially every informative day at both grids and no inference ships for it; the contradiction is unadjudicated (F-4). E-3 · *half_life_s* in the ECM-SDE curves is in bar units; the label is wrong by $\times 100$ at 10 ms (F-14). E-4 · The shipped FEVD/GFEVD CSVs are the benchmark-regime tables, unlabeled; volatile-regime decompositions exist only inline in report.md. E-5 · Rigobon verdict annotation: the eigenvalue-gap row passes its own printed threshold while the verdict is NO (M-4). E-6 · Copula "median $p(\text{upper-tail LR}) = 0.0$ " in all 8 groups, including groups whose winning family has zero tail dependence by construction; numerically suspect, do not quote. E-7 · Table 5's two independence nulls differ within each panel (unexplained asymmetric benchmark); Table 7's 1 s ratio (33.54) does not reproduce from its own rounded columns (33.94); "NCMOF" denotes different objects in Tables 5 and 7. E-8 · Regime taxonomy differs across table families (M-8); tier-1 MWCB bar counts differ by one across grids (1501/1502); *pd_link* window counts 852 vs 849. E-9 · The 10 ms pooled panel-VECM regime split (benchmark IS 0.116 / volatile 0.574) is far outside the per-day IS means (0.437 / 0.484) under the same heading, unreconciled. E-10 · The MWCB duration-asymmetry p is bit-identical across grids (0.1263436828158592) while every input differs — shared permutation stream; verify rather than assume coincidence. E-11 · *t_delta_es* in the 10 ms liquidity-state tables disagrees in sign with the printed q_{10} – q_{90} spread on 22/66 (depth) and 15/66 (FPCA) days; the column's tested quantity

is undefined in the bundle. E-12 · Single-session exhibits (*_session0, liquidity curves) all use 2017-06-26 — a benchmark day with the third-lowest ES update count, the ladder-monotonicity defect (D-3), and the largest pre-averaging outlier; it is unrepresentative in the directions that matter. E-13 · frames/report.md/summary.json are zero-stage placeholders ([null, null] headline) from the frames-build pass; readable as a failure though they are not one. E-14 · The QC narrative asserts futures price limits are reported while its own coverage column reads 0 % on 65/66 sessions (D-8). E-15 · Lag bookkeeping: p = 10 (STAGE 4c) vs p = 5 (Table 9 frames) vs n_lags 5/60 (VECM stages) vs modal per-day p = 1 (M-3).

VI. Disposition and Remedies

1. **Adopt at 1 s, flag grid-dependence.** F-1 (with its conservative-bias support), F-6, F-7's liquidity-split contrast, F-8, F-9, F-13 are cross-checked and quotable now. F-2, F-3's suppression structure, F-11's duration asymmetry, and F-12 are 1 s-grid results and must be presented as such.
 2. **v0.9.87 (shipped) addresses:** M-2 (Table 5 day-clustered inference + within-pair contrast), the within-pair IS test (F-5), M-3 (lag-free LP Table 9), F-7's t-family constraint (nonparametric λ columns), D-1/D-2 (measured bias curve; es_stale_frac per-day covariate), D-5 (es_contract QC column), and the tick-regime controls (per-day control standardization; ex-straddle within-pair rows).
 3. **Next run configuration:** PEAK_GB_GUESS=381; rerun the 10 ms Table 9 with the RealBar column skipped (informationally identical to 1 s at 60 s bars) and n_boot = 99; set mediation n_boot = 2000 with BCa; raise MS-regime draws if the p-floor is to carry information; run the Table 9 --lag-band; run T9_LP=1.
 4. **Still open after v0.9.87:** a formal MS-state/label concordance statistic; adjudication of raw-vs-pre-averaged θ (E-2); the ECM-SDE half-life unit label (E-3); labeling of the FEVD/GFEVD regime CSVs (E-4); the copula upper-tail LR column (E-6); the t_delta_es definition (E-11); replacement of 2017-06-26 as the session0 illustration (E-12).
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VII. Verification Statement

This report was compiled from a parallel reading of all 132 bundle files by eight independent readers with forced structured output, followed by adversarial re-verification of all 60 headline and data-quality findings against their cited source files. Fifty-nine verified verbatim; one (the extraction worker count) was corrected from 5 to 4. Two files initially uncovered (the 10 ms liquidity-conditional state tables) were read in a completeness pass and are incorporated (E-11). All figures quoted above are verbatim from the cited tables or arithmetic recompositions of them; recomputed quantities (regime splits of per-day tables, staleness thresholds, correlation of lead-lag with update counts) are identified as such in the underlying record.